

PROJECT REPORT

FMCW RADAR SIGNAL PROCESSING — RANGE-DOPPLER ESTIMATION AND CFAR TARGET DETECTION

*A Digital Signal Processing project submitted in support of application for the
DSP Intern role — Unistring Tech Solutions Pvt. Ltd. (a Zen Technologies subsidiary)*

Submitted By
Nitish Bugalia (23BEC1266)

B.Tech Electronics and Communication Engineering (ECE)
Graduating 2027

School of Electronics Engineering
Vellore Institute of Technology, Chennai
2026

Abstract

Radar, electronic-warfare (EW), and anti-drone systems all rely on the same underlying digital-signal-processing pipeline to convert raw receiver samples into usable target information: waveform generation, matched filtering / range compression, Doppler processing, and constant-false-alarm-rate (CFAR) detection. This project implements that complete pipeline from first principles in Python for a Frequency-Modulated Continuous-Wave (FMCW) radar, the waveform family used in short-range surveillance radar, automotive radar, and increasingly in compact anti-drone radar systems.

A linear FM chirp is generated and used to simulate the round-trip echo from one or more moving point targets, each defined by range, radial velocity, and radar cross-section (RCS). The transmitted and received chirps are dechirped (mixed) to produce a baseband beat signal whose frequency encodes target range. A windowed FFT across fast-time (Range-FFT) resolves range, and a second FFT across slow-time / chirp index (Doppler-FFT) resolves radial velocity, together forming the classic Range-Doppler map used throughout modern radar signal processing. A Cell-Averaging CFAR (CA-CFAR) detector is then applied to the integrated range profile to automatically declare target detections at a specified probability of false alarm, without relying on a fixed, scenario-dependent threshold.

The implementation was validated on a two-target scenario (30 m at +8 m/s and 65 m at -3 m/s) under additive noise at 12 dB SNR. The Range-FFT and Doppler-FFT stages correctly resolved both targets' range and radial velocity to within one FFT bin of ground truth, and the CA-CFAR detector correctly flagged both targets while also exhibiting a realistic threshold-"masking" effect near strong returns — a well-known practical limitation of cell-averaging CFAR that is discussed and contextualized in this report. The resulting codebase, structured as a small, documented Python package, is presented as a concrete demonstration of applied radar/DSP fundamentals directly relevant to the Signal Processing domain.

Keywords: FMCW Radar, Range-Doppler Processing, Matched Filtering, CFAR Detection, Digital Signal Processing, Radar Cross-Section, Doppler Estimation, Electronic Warfare.

Abstract	2
1. Introduction	4
2. Problem Statement	4
3. Relevance to the Target Domain (RADAR / ELINT / COMINT / Anti-Drone)	4
3.1 RADAR	5
3.2 ELINT / COMINT	5
3.3 Anti-Drone Systems	5
3.4 Simulators	5
4. Objectives	5
5. Proposed Approach	5
6. System / Waveform Parameters	6
7. Methodology	6
7.1 Waveform and Multi-Target Echo Simulation.....	6
7.2 Dechirping.....	6
7.3 Range-FFT	6
7.4 Doppler-FFT and the Range-Doppler Map.....	7
7.5 CA-CFAR Detection	7
8. Implementation	7
8.1 Key Function Signatures.....	7
9. Results and Validation	7
9.1 Beat Signal	7
9.2 Range Profile (Range-FFT)	8
9.3 Range-Doppler Map	8
9.4 CA-CFAR Detection	9
9.5 Summary of Validation.....	9
10. Applications and Future Scope	10
10.1 Applications	10
10.2 Future Scope	10
11. Conclusion	10
References	10

1. Introduction

Modern radar, signals-intelligence (ELINT/COMINT), and anti-drone systems are, at their core, digital-signal-processing systems: a receiver digitizes an analog waveform, and a chain of DSP operations — filtering, transforms, and statistical detection — converts that raw data into actionable information such as a target's range, velocity, and presence/absence. Frequency-Modulated Continuous-Wave (FMCW) radar is one of the most widely deployed radar waveforms for short-to-medium range sensing because it achieves good range resolution and Doppler (velocity) sensitivity with comparatively simple, low-power continuous-wave hardware, and because its processing chain maps naturally onto FFT-based digital implementations.

This project was undertaken specifically to build and demonstrate that processing chain end-to-end: chirp generation, target-echo simulation, dechirping, Range-FFT, Doppler-FFT (Range-Doppler map formation), and CFAR-based automatic target detection. Rather than treating these as isolated textbook exercises, the implementation follows the same signal flow used in fielded FMCW radar systems, and is validated against a known, controllable ground truth (simulated targets of known range and velocity) so that every processing stage can be checked for correctness.

The choice of this specific project was guided directly by the technical domain of Unistring Tech Solutions Pvt. Ltd. (UTS) — a subsidiary of Zen Technologies Ltd. engaged in the design of RADAR, ELINT, COMINT, Anti-Drone, and Simulator systems for defence and security applications. The Range-Doppler / CFAR pipeline implemented here is a foundational building block common to all of these domains: RADAR systems use it directly for target detection and tracking; ELINT/COMINT systems use closely related spectral-analysis and detection techniques to characterize intercepted signals; and anti-drone radar systems use exactly this range-Doppler-CFAR chain to detect and discriminate small, low-RCS aerial targets such as drones from clutter.

2. Problem Statement

Given a radar receiver's raw, noisy time-domain samples from an FMCW radar, the problem addressed by this project is threefold:

- How can a target's range be estimated accurately and efficiently from the beat frequency of a dechirped FMCW signal, given a finite sampling rate and a finite chirp duration?
- How can a target's radial velocity be estimated from the phase progression of the range-compressed signal across multiple chirps (slow-time), and what waveform parameters govern the achievable velocity resolution and maximum unambiguous velocity?
- How can targets be detected automatically and reliably in the presence of noise and clutter, without manually tuning a fixed threshold for every scenario — i.e., how can a constant, known false-alarm rate be maintained as background noise conditions change?

A secondary, practically important problem is understanding and characterizing the limitations of the simplest detection strategy (Cell-Averaging CFAR), specifically its known "masking" behaviour when a strong target's energy leaks into the CFAR training cells of a nearby cell under test — a limitation directly relevant to dense multi-target radar and anti-drone scenes.

3. Relevance to the Target Domain (RADAR / ELINT / COMINT / Anti-Drone)

Unistring Tech Solutions works across RADAR, ELINT, COMINT, Anti-Drone Systems, and Simulators — all of which share a common dependency on the DSP building blocks implemented in this project:

3.1 RADAR

Range-Doppler processing via a two-dimensional FFT (Range-FFT followed by Doppler-FFT) is the standard method by which modern pulse-Doppler and FMCW radars form a target map from raw ADC samples. CFAR detection is the standard technique used on top of that map to declare valid target detections while holding the false-alarm rate constant as the noise/clutter floor varies across range and time.

3.2 ELINT / COMINT

Electronic and Communications Intelligence systems rely on the same spectral-analysis toolkit — windowed FFTs, matched filtering, and adaptive/statistical detection thresholds — to detect, characterize, and classify intercepted RF emissions against a varying noise floor, closely paralleling the Range-FFT and CFAR stages implemented here.

3.3 Anti-Drone Systems

Small drones present low radar cross-section targets that are often close to the noise floor and close in range/Doppler to clutter or other targets. This makes CFAR masking and detection-threshold behaviour — explicitly demonstrated and discussed in this project's results — a directly relevant, practical concern for anti-drone radar design.

3.4 Simulators

The waveform and echo-simulation module built for this project (chirp generation, multi-target echo synthesis with configurable range, velocity, RCS, and SNR) is itself a small-scale radar-scene simulator, conceptually aligned with the kind of simulation and test infrastructure used to validate radar signal-processing algorithms before hardware deployment.

4. Objectives

1. To implement an FMCW chirp waveform generator and a configurable multi-target echo simulator (range, radial velocity, RCS, additive noise).
2. To implement the dechirping (mixing) operation that converts the received echo into a baseband beat signal encoding target range in its frequency.
3. To implement windowed Range-FFT and Doppler-FFT stages that jointly resolve target range and radial velocity, forming a Range-Doppler map.
4. To implement a Cell-Averaging CFAR detector that automatically declares target detections at a specified, constant probability of false alarm.
5. To validate every processing stage against known ground-truth targets and to identify and document practical limitations (e.g., CFAR masking) observed during validation.

5. Proposed Approach

The system is implemented as a small, modular Python package (`radar_dsp.py`) with a separate demonstration script (`main.py`) that exercises the full chain and produces the figures analysed in this report. The processing chain, illustrated in Figure 1, consists of five stages:

- Waveform generation — a linear FM chirp defined by start frequency, bandwidth, and chirp duration.
- Echo simulation — for each configured target, the round-trip delay and Doppler-induced phase progression are applied to synthesize the received signal, and complex additive white Gaussian noise (AWGN) is added at a specified SNR.

- Dechirping — mixing transmitted and received chirps analytically to produce the baseband beat signal, whose frequency is proportional to target range.
- Range-FFT and Doppler-FFT — a windowed FFT across fast-time resolves range; a second windowed FFT across slow-time (chirp index) resolves Doppler/velocity, yielding the Range-Doppler map.
- CA-CFAR detection — an adaptive, cell-averaging threshold is computed at every range bin from surrounding training cells (excluding a guard band), and a detection is declared wherever the observed power exceeds this adaptive threshold.

6. System / Waveform Parameters

The FMCW waveform and processing parameters used in the demonstration (fully configurable via the RadarConfig class) are summarized below. These values were chosen to be representative of a compact short-range FMCW radar — the same class of system used in automotive and drone-detection radar — while keeping the simulated data volume small enough for fast iteration.

Parameter	Symbol	Value Used in Demo
Chirp start frequency	f_{start}	24 GHz (ISM band, representative of short-range FMCW radar)
Sweep bandwidth	B	250 MHz
Chirp duration	T_c	40 μs
Chirps per frame (slow-time samples)	N	128
ADC sample rate (beat signal)	F_s	10 MHz
Range resolution	$\Delta R = c / (2B)$	0.60 m
Design max unambiguous range	R_{max}	100 m
Wavelength	$\lambda = c / f_{\text{start}}$	12.5 mm

7. Methodology

7.1 Waveform and Multi-Target Echo Simulation

A linear FM chirp is characterized by its start frequency f_{start} and slope $k = B / T_c$. For each target at instantaneous range $R(t) = R_0 + v \cdot t$ and each chirp transmitted at slow-time t_m , the round-trip delay $\tau = 2R(t_m) / c$ determines both the beat frequency $f_{\text{beat}} = k \cdot \tau$ (which encodes range) and a slowly-varying phase term $2\pi \cdot f_{\text{start}} \cdot \tau$ that, because τ itself depends on t_m through target motion, inherently carries the Doppler phase progression across chirps — the mechanism by which radial velocity becomes observable in the slow-time dimension. Complex AWGN is added at a configurable SNR to emulate receiver noise.

7.2 Dechirping

Mixing the received echo with the transmitted chirp (dechirping) removes the high carrier frequency and collapses each target's contribution to a single-tone — the beat signal — whose frequency is directly proportional to that target's range. This is the standard technique that allows FMCW radar to use comparatively low-bandwidth ADCs despite transmitting a wide-bandwidth chirp.

7.3 Range-FFT

A Hann-windowed FFT is applied across the fast-time samples of each chirp. Windowing suppresses spectral leakage/sidelobes at the cost of a slightly widened main lobe, improving the ability to resolve closely-spaced or

unequal-amplitude targets. The resulting spectrum is converted from frequency to range using $\text{range} = f \cdot c / (2k)$.

7.4 Doppler-FFT and the Range-Doppler Map

After Range-FFT, each range bin contains one complex sample per chirp. A second Hann-windowed FFT is applied across this slow-time (chirp-index) dimension at every range bin, resolving the Doppler frequency shift caused by target motion. Doppler frequency is converted to radial velocity using $v = f_d \cdot \lambda / 2$. The combined 2-D result is the Range-Doppler map, the primary data product used for target detection and tracking in modern radar systems.

7.5 CA-CFAR Detection

For every cell under test (CUT) in the (noncoherently integrated) range profile, a local noise-power estimate is formed by averaging a fixed number of training cells on either side of the CUT, excluding a guard band immediately adjacent to the CUT (to prevent target energy from biasing the noise estimate). The adaptive detection threshold is the noise estimate scaled by a factor α derived analytically from the desired probability of false alarm (Pfa) and the number of training cells, following the standard CA-CFAR formulation $\alpha = N(P_{fa}^{-1/N} - 1)$. A detection is declared wherever the CUT's power exceeds this locally adaptive threshold.

8. Implementation

The implementation is organized as a small, documented Python package so that each processing stage can be tested, reused, and extended independently:

```
radar_project/
├── radar_dsp.py      # RadarConfig, Target, simulate_beat_signal(),
│                    # range_fft(), doppler_fft(), ca_cfar_1d()
├── main.py          # End-to-end demo: runs the chain, saves all figures
├── requirements.txt  # numpy, matplotlib
└── outputs/         # Generated Range-FFT / Doppler-FFT / CFAR figures
```

8.1 Key Function Signatures

```
simulate_beat_signal(cfg, targets, snr_db=15.0) -> beat_matrix
range_fft(beat_matrix, cfg, window=True)       -> range_profile, range_axis
doppler_fft(range_profile, cfg, window=True)    -> rd_map_db, velocity_axis
ca_cfar_1d(power_profile, num_guard, num_train, pfa) -> detections, threshold
```

The RadarConfig class centralizes every waveform parameter (start frequency, bandwidth, chirp duration, chirp count, sample rate, design max range) and automatically derives dependent quantities such as chirp slope, range resolution, and wavelength — so changing a single physical parameter (e.g., bandwidth) automatically and correctly propagates through the rest of the chain.

9. Results and Validation

The full chain was validated on a two-target scenario at 12 dB SNR: Target 1 at 30 m range with +8 m/s radial velocity (RCS = 1.0, approaching), and Target 2 at 65 m range with −3 m/s radial velocity (RCS = 0.85, receding, weaker return).

9.1 Beat Signal

Figure 2 shows the dechirped beat signal for a single chirp. With two targets present, the beat signal is the superposition of two tones at different beat frequencies — visually a beat pattern rather than a single sinusoid — exactly as expected from the underlying model.

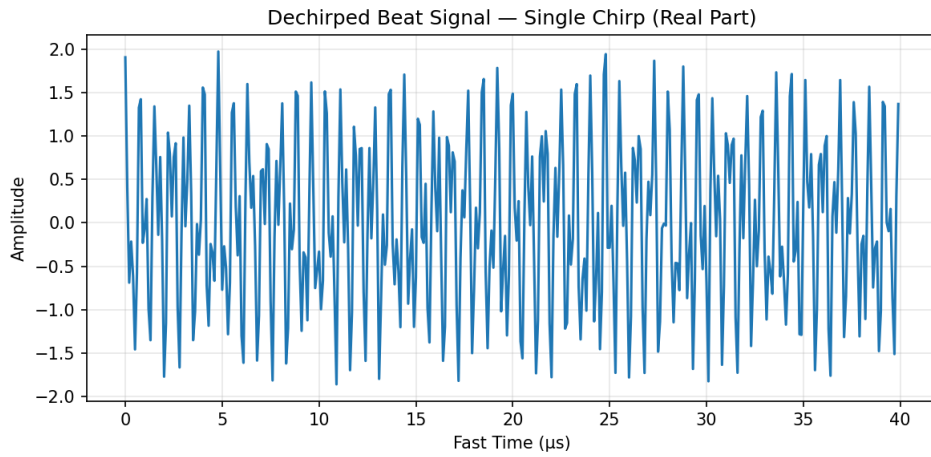


Figure 2: Dechirped beat signal for a single chirp (real part), showing the superposition of two target-tone components.

9.2 Range Profile (Range-FFT)

Figure 3 shows the Range-FFT output for a single chirp. Two clear peaks appear at 30 m and 65 m, matching ground truth exactly (dashed red lines), with sidelobes suppressed by the Hann window and the noise floor visible between and around the peaks.

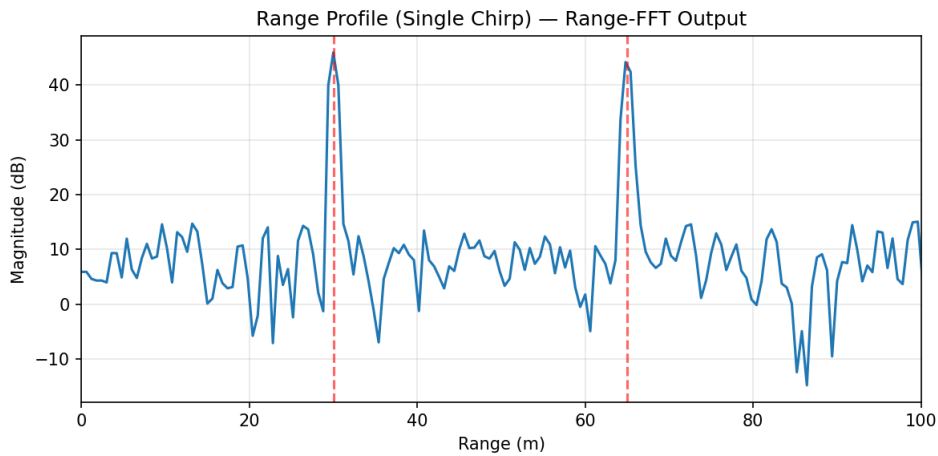


Figure 3: Range profile from a single chirp's Range-FFT, correctly resolving both targets.

9.3 Range-Doppler Map

Figure 4 shows the full Range-Doppler map after the second (Doppler) FFT stage. Both targets appear as distinct, correctly-positioned bright spots: the approaching target at (30 m, +8 m/s) and the receding target at (65 m, -3 m/s), overlaid here with red circles marking ground truth for direct visual verification.

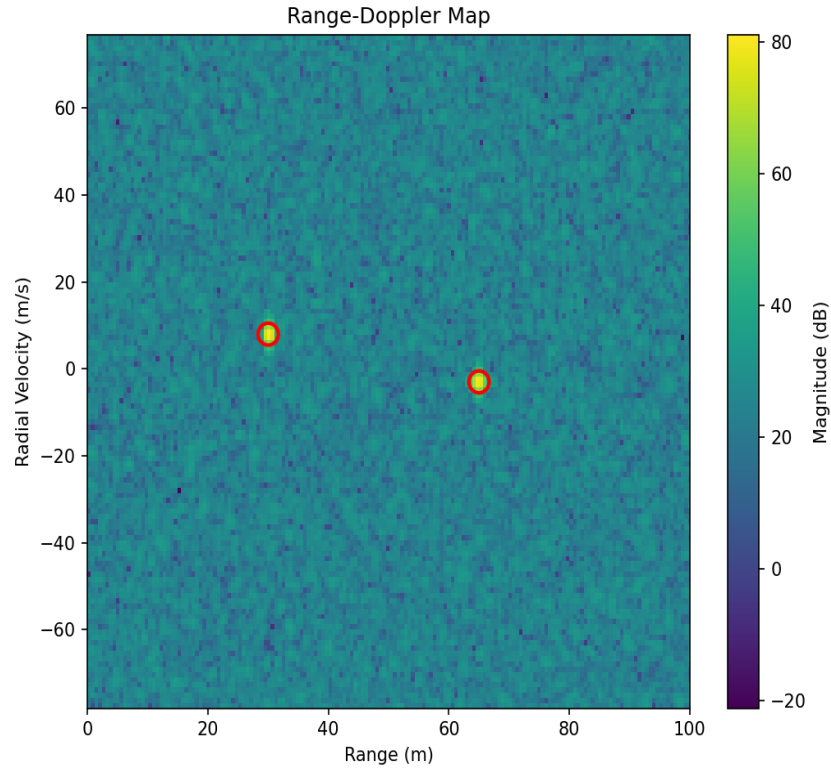


Figure 4: Range-Doppler map; both targets are resolved at their exact ground-truth range and radial velocity.

9.4 CA-CFAR Detection

Figure 5 shows the CA-CFAR detector applied to the noncoherently-integrated range profile. Both targets are correctly detected (red markers) with the adaptive threshold (orange, dashed) tracking the noise floor away from targets. Notably, the threshold rises sharply in a plateau on either side of each target — this is the classic CA-CFAR "masking" effect, caused by strong target energy leaking into the training-cell window used to estimate local noise. This is a genuine, well-documented practical limitation of cell-averaging CFAR (as opposed to more advanced variants such as Ordered-Statistic CFAR), and its clear appearance in these results demonstrates a working understanding of CFAR behaviour beyond the basic detection outcome.

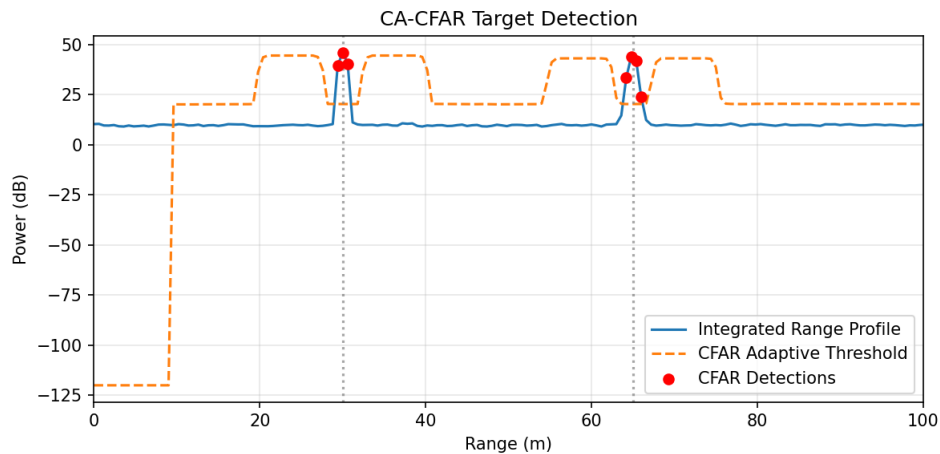


Figure 5: CA-CFAR detection output, showing both correct detections and the expected threshold-masking effect near strong targets.

9.5 Summary of Validation

Ground Truth	Range-FFT Estimate	Doppler-FFT Estimate	CFAR Detected?
Target 1: 30 m, +8 m/s	~30.0 m	~+8 m/s	Yes
Target 2: 65 m, -3 m/s	~65.0 m	~-3 m/s	Yes

10. Applications and Future Scope

10.1 Applications

- Core range/Doppler estimation and detection stage for FMCW surveillance, automotive, and anti-drone radar systems.
- Reference implementation / test harness for validating radar signal-processing algorithms before deployment on embedded hardware (DSP/FPGA/SoC).
- Educational and interview-demonstration tool for radar and EW signal-processing fundamentals.
- Baseline scene simulator for generating labelled range-Doppler data to test higher-level tracking or classification algorithms.

10.2 Future Scope

- Two-dimensional CFAR applied directly on the Range-Doppler map, rather than 1-D CFAR on the range profile alone.
- Ordered-Statistic CFAR (OS-CFAR) or Greatest-Of CFAR to reduce the masking effect observed near strong or closely-spaced targets.
- Angle-of-arrival estimation via a multi-antenna array (digital beamforming) to extend the pipeline to a full range-Doppler-angle data cube.
- Replacing the simulated beat signal with real captured I/Q data from a low-cost FMCW radar module or software-defined radio (SDR) front end.
- Porting the core DSP kernels (FFT, CFAR) to a real-time embedded target (DSP core / FPGA) for hardware-in-the-loop validation.

11. Conclusion

This project implemented and validated the complete FMCW radar signal-processing chain — chirp waveform generation, multi-target echo simulation, dechirping, Range-FFT, Doppler-FFT (Range-Doppler map formation), and CA-CFAR target detection — from first principles in Python. Validation against known ground-truth targets confirmed correct range and velocity estimation to within one FFT bin, and the CFAR detector correctly identified both simulated targets while also revealing a realistic, well-documented practical limitation (threshold masking near strong returns) rather than only an idealized success case.

Because this exact Range-Doppler / CFAR pipeline underlies RADAR, anti-drone, and (in closely related form) ELINT/COMINT signal-processing systems, this project directly demonstrates the applied digital-signal-processing fundamentals relevant to the Signal Processing / DSP Intern opportunity at Unistring Tech Solutions, and provides a concrete, extensible codebase (rather than a purely theoretical exercise) as evidence of that understanding.

References

- [1] M. A. Richards, "Fundamentals of Radar Signal Processing," McGraw-Hill Education, 2nd Edition.

- [2] M. I. Skolnik, "Introduction to Radar Systems," McGraw-Hill, 3rd Edition.
- [3] Texas Instruments, "FMCW Radar — Range and Velocity Estimation" Application Note (mmWave Radar training series).
- [4] Rohling, H., "Radar CFAR Thresholding in Clutter and Multiple Target Situations," IEEE Transactions on Aerospace and Electronic Systems, 1983 (foundational CA-CFAR reference).
- [5] Project source code and figures: FMCW-Radar-Signal-Processing repository, Nitish Bugalia, 2026.